SUDHEESH P

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EDUCATION

Government College of Engineering, Trivandrum (CET) 2025 – 2028
B.Tech in Electrical and Computer Engineering
Minor in Artificial Intelligence

Diploma in Computer Engineering 2023 – 2025
IPT & GPTC Shoranur

EXPERIENCE

Research Intern – Quantum Systems & Computational Microwave Imaging, *CET* *May – July 2026*
Conducted research under Dr. Kiran R. (Department of Electronics and Communication Engineering) on the coupled Maxwell–Schrödinger framework, simulating the interaction between classical electromagnetic fields and quantum mechanical systems in nanotechnology. Designed and implemented quantum algorithms executed on IBM's large-scale quantum computing infrastructure, and applied AI/ML techniques for quantum state analysis, simulation optimization, and computational microwave imaging investigations.

G-Tec, *MERN Intern* *April 2024 - May 2024*
Built and maintained full-stack web application projects using MongoDB, Express.js, React, and Node.js, implementing RESTful APIs for seamless frontend-backend communication.

International Conclave on Generative AI and the Future of Education 2.0 *December 2024*
Engaged with global experts and innovators discussing the transformative impact of generative AI on education, along with workshops, a hackathon, and a public exhibition. The event was co-organized by the Higher Education Department of the Government of Kerala and the Institute of Human Resources Development (IHRD).

PROJECTS

Mars Rover Dust Mitigation — *Ongoing*
• Working on a mission study assigned by the Vikram Sarabhai Space Centre (VSSC) under Prof. Murali Krishnan K H and Dr. Sreeja S.
• Designing systems to mitigate dust deposition on the solar panels of the rover proposed under Indian Space Research Organisation's (ISRO) Mangalyaan-2 mission, safeguarding power generation across a long-duration surface mission in the Martian dust environment.

Vespera — *overview*
• Architected an AI-driven healthcare orchestration platform powered by NEHA (Next-generation Environment for Human Advancement), integrating appointment scheduling, predictive hospital availability, dynamic queue tracking, transport logistics (cab/ambulance coordination), medicine delivery, and longitudinal medical-record management via privacy-preserving Universal and Temporary Vespera IDs.
• Designed a unified patient interface enabling prescription management, lab integration, home-consultation scheduling, real-time care navigation, and secure, time-limited clinical data sharing with providers and pharmacies.
• Developed physician-assistive intelligence including ambient consultation capture for auto-generated clinical documentation, one-action administrative automation ("Magic Button"), and real-time decision-support insights ("Magic Whispers") derived from multimodal patient data.
• Engineered simulation-driven clinical tools: Digital Twin models for pre-intervention testing, Time Machine for longitudinal outcome forecasting and protocol comparison, and HoloPatient mixed-reality visualization for procedure validation with quantified risk and outcome prediction.
• Extended system capabilities toward population-scale analytics and early-warning detection using anonymized health signals to model disease trends and support proactive public-health response.

Spiritus https://spiritus.vercel.app/ — *React.js, Node.js, Express, MongoDB, Gemini-3-pro*
• Spiritus offers direct access to a global network of clinicians, psychotherapists, and licensed mental health professionals, delivering tailored, evidence-based support in various languages for your personal transformation and the pursuit of a meaningful existence.
• We provide online therapy for depression, anxiety, PTSD, stress-related burnout, chronic loneliness, and insomnia, offering compassionate, accessible care to individuals, couples, and teenagers worldwide.

Starbase https://starbase-x.vercel.app/ — *DBMS*

• Developed a database-driven material selection recommendation system that helps students, engineers, designers and researchers identify the most suitable engineering material for an application, replacing manual searching through material handbooks.
• Modelled a structured database of metals, polymers, ceramics, and composites, queried against user requirements: application, operating conditions, load, temperature range, corrosion resistance, weight, manufacturing process, and budget.
• Returned the best-matching materials with their key properties and justifications, using predefined selection criteria and database queries.
• Designed around a centralised database for data consistency, easy maintenance, and scalability, so new materials and properties can be added with minimal effort.

Coastal Adviser https://costal-adviser.vercel.app/ — *FastAPI, React.js, Playwright, Sarvam AI*

• Architected an AI-powered decision-support platform for Kerala's coastal fishing community, integrating real-time maritime safety, market intelligence, and government welfare access.
• Engineered an automated maritime data pipeline using Playwright to scrape and process INCOIS advisories (Potential Fishing Zones, wave height, wind speed) for 20+ coastal locations.
• Developed a market price tracking system with 7-day trend analysis for major Kerala fish markets, providing data-driven insights to advise fishermen on optimal selling vs. cold-storage strategies.
• Implemented a low-latency voice-to-voice interface using Sarvam AI and OpenAI, delivering multilingual support in Malayalam and English to ensure accessibility for non-literate users.
• Designed a lightweight, zero-dependency Retrieval-Augmented Generation (RAG) system for semantic search across state and central government welfare schemes, optimizing for local context.
• Built a hyperlocal personalization engine using geofencing and user profiles (boat type, preferred species) to deliver targeted maritime alerts and species-specific economic intelligence.

Socius

• Designed a mobile-first hyperlocal platform enabling short-term borrowing of everyday goods within verified communities, digitising informal neighborly sharing to reduce unnecessary purchases and promote sustainable access over ownership.
• Implemented geofenced discovery across apartments, campuses, and neighbourhood clusters, allowing users to locate, request, and exchange items within walkable proximity without logistics or shipping overhead.
• Engineered a secure refundable-deposit (escrow-style) mechanism to ensure accountability—protecting lenders against damage or loss while keeping borrowing near-free and non-commercial.
• Built trust infrastructure through multi-factor user verification (identity, phone, and residence validation), reputation scoring, reviews, and gamified "karma" incentives to encourage reciprocal participation.
• Developed automated workflows including searchable community inventories, booking management, reminders, digital agreements, and dispute-resolution tooling to streamline peer-to-peer exchanges.
• Positioned the system as a circular-economy enabler, extending product lifecycles, reducing household expenditure, and strengthening local social networks while adapting the global "Library of Things" model to Indian urban dynamics.

Genus *Flutter, Firebase*

• A cross-platform pet marketplace enabling seamless connections between buyers, sellers, and pet enthusiasts through intelligent matching algorithms and geolocation-based discovery.
• Built a secure authentication system with role-based access control, integrated payment processing, and real-time messaging to facilitate trusted transactions and community engagement.

ACHIEVEMENTS

Code reCET 2.0 Project Expo, *First Prize* *2025*

Awarded First Prize for presenting the Spiritus project at the Code reCET 2.0 Project Expo held at the College of Engineering Trivandrum (CET). The expo offered a distinguished platform to showcase innovative work before a tech-savvy audience comprising industry professionals, peers, and mentors.

InnoFEST25 Project Expo, *First Prize* *2025*

Honored with First Prize for the "Spiritus" project at InnoFEST25, hosted by the Innovation and Entrepreneurship Development Centre (IEDC) and the Institution's Innovation Council (IIC) under the Ministry of HRD.

RESEARCH

Meshless Radial Point Interpolation for Self-Consistent Maxwell–Schrödinger Modelling of Nanoscale Devices

Science Undergraduate Research Conference 2026, Azim Premji University *5 December 2026*
Presenting research on a meshless radial point interpolation method (RPIM) for the coupled Maxwell–Schrödinger modelling of classical electromagnetic fields and quantum-mechanical systems in nanoscale devices, building on an FDTD study of the semiclassical framework (potential-based formulation, Lorenz gauge, PML boundaries).

Plenum Theory: On the Origin of Spacetime *Manuscript in Preparation, 2026*
Independent theoretical physics research proposing a unified algebraic framework for spacetime, the Standard Model, and cosmological evolution. Selected contributions include: a theorem forcing $d = 4$ from algebraic consistency alone (zero free parameters); a proof that fermion generations number exactly three as a structural necessity of the underlying algebra; zero-parameter derivations of inflationary observables matching Planck 2018 to 0.35σ; and a dark energy prediction matching DESI DR2 (2025) to 0.15σ from a single dynamical mechanism. The manuscript contains 55 theorems and derives the Standard Model gauge structure, mixing hierarchies, and cosmological observables from a single algebraic postulate.

SKILLS

Languages	C, Python, MATLAB, JavaScript, Dart, Tailwind CSS
Technologies	React, Node.js, Flutter, Django
Tools	Git, Postman, Linux, HPC, LaTeX
Database	MongoDB, MySQL
Communication	English, Spanish, Sanskrit, Malayalam

SOCIAL MEDIA

LinkedIn	linkedin.com/in/sudheeshpoolakkal/